

Using Explainable Techniques to Enhance Chain of Thoughts in LLMs

1st Nafis Tanveer Islam

Multiscale Networked Systems (MNS)
University of Amsterdam
n.t.islam@uva.nl

2nd Ruoyu Wang

Multiscale Networked Systems (MNS)
University of Amsterdam
ruoyu.wang2@student.uva.nl

3rd Zhiming Zhao *

Multiscale Networked Systems (MNS)
University of Amsterdam
z.zhao@uva.nl

Abstract—With the recent advancements in reasoning capabilities with semantic understanding, Large Language Models (LLMs) are increasingly mimicking human reasoning processes and aligning more closely with human values. Yet, unlike how humans sometimes make decisions intuitively by reasoning or understanding the underlying logic, LLMs struggle with transparency in their internal reasoning. Similar to how humans often pause to analyze their reasoning step-by-step to understand their own decisions better, we propose enabling LLMs to perform analogous introspective reasoning using a Chain-of-Thought (CoT) framework powered by SHAP-based explainability. CoT method guides LLMs to explicitly render intermediate reasoning steps, effectively simulating a human’s reflective thought process. Meanwhile, SHAP explainability provides context or logic by quantifying how each semantic token or feature contributes to the reasoning, analogous to a human examining which factors most influenced their choices. Our analysis explores how various training methods influence their semantic comprehension and, consequently, their ability to communicate their reasoning transparently. We apply this approach to a specialized environmental and Earth science ranking dataset—characterized by user feedback and sparse annotations—and we assess whether LLM-generated rankings can not only reflect accurate semantic understanding but also transparently emulate human-like introspection.

Index Terms—LLM, Explainability, Reasoning, Chain-of-Thought (CoT).

I. INTRODUCTION

With the ongoing advancements in semantic understanding, Large Language Models (LLMs) increasingly demonstrate human-like reasoning patterns, thus aligning with human-like decision-making processes. In knowledge retrieval systems, an improved semantic capability for ranking can substantially enhance the user’s access to relevant information [2]. However, despite their enhanced accuracy, traditional re-ranking systems often lack transparency, providing little insight into the reasoning behind their ranking decisions.

Much like humans who occasionally rely on intuition but often pause to reflect and analyze their decisions explicitly, current LLMs tend to produce outcomes without clearly articulating the underlying logic. The use of traditional recommendation strategies, including Collaborative Filtering (CF) lacks transparency, providing generic rationales like “other users liked this” rather than providing an item-specific reasoning

[3]. This limits user trust and the perceived reliability of recommendations [4].

To address this challenge, we put forth a comprehensive reasoning approach where LLMs mimic a human cognitive process of explicitly reflecting on decisions. Therefore, we manipulate the Chain-of-Thought (CoT) prompting technique, enabling the LLM to articulate intermediate reasoning steps sequentially. This step-by-step introspection makes the semantic logic behind rankings transparent and understandable, mirroring how humans consciously reflect to justify their choices. We evaluate our introspective CoT and SHAP-driven approach using a specialized dataset from ENVRI-Hub-Next¹, tailored explicitly for knowledge sharing among environmental and Earth science domains [1].

II. SYSTEM DESIGN AND EXPERIMENTS

A. Methodology

To optimize LLM ranking quality, we explore several training strategies, namely Supervised Fine Tuning (SFT), Proximal Policy Optimization (PPO), Direct Policy Optimization (DPO), and Reward Modeling—using explicit query-item ranking pairs. After training, we reduce the model to a single output neuron, which assigns a relevance score to each candidate for re-ranking. Finally, we use the trained model with SHAP explainability to generate attribution scores for each token. Then we pick the top ten tokens with the highest attribution scores, and pair them with a prompt so that the LLM can reason based on its understanding using the tokens.

B. Dataset Collection

The dataset was collected during the ENVRI-Hub-Next Hackathon, where 12 domain experts queried the ENVRI-Hub Knowledge Base and re-ranked the top 10 BM25-retrieved results per query on a scale from 0 (irrelevant) to 9 (highly relevant). These expert annotations form the supervision data for training and evaluating our re-ranking models. We used Direct Policy Optimization (DPO) for training. We split the top five-ranked responses as positives and the bottom five as negatives, creating 25 pairwise comparisons per query, yielding 2,350 training pairs in total.

* Corresponding Author

¹<https://envri.eu/envri-hub-next/>

TABLE I
COMPARISON OF RANKING PERFORMANCE OF VARIOUS MODELS

Model	NDCG@5	Hits@5	NDCG@10	Hits@10
BM25	0.6434	0.70	0.7193	0.70
LoRA-DPO	0.5470	0.60	0.6592	0.70
Llama-2.7B-hf				
Qwen	0.6657	0.60	0.7314	0.70
Reward-model	0.5905	0.60	0.6774	0.70
Llama 3.1				
Human	0.8000	0.70	0.8000	0.70

C. Evaluation of Ranking Models

Table I summarizes our offline evaluation of different ranking models on *NDCG@5*, *Hits@5*, *NDCG@10*, and *Hits@10* metrics. Table I presents the ranking performance of several models, including classical baselines, LLM-based approaches, and human-annotated ground truth. Among the automated systems, the Qwen achieves the highest ranking effectiveness, outperforming all others in both *NDCG@5* and *NDCG@10*. Notably, Qwen attains an *NDCG@10* of 0.7314, closely approaching human performance (0.8) and demonstrating strong alignment with relevance-based ranking expectations.

In contrast, the LoRA-DPO variant of LLaMA-2.7B underperforms across all metrics, suggesting that lightweight parameter-efficient fine-tuning may not be sufficient for capturing nuanced semantic relationships in re-ranking tasks. The Reward-model LLaMA 3.1 performs moderately, improving over LoRA-DPO but still falling short of Qwen. Interestingly, BM25 remains competitive in terms of Hits5 and Hits10 (both at 0.70), showing its strength in retrieving relevant items.

Initial Query: “Temperature in the Alps in 2022”

Top Search Result Types: Climatic reports, station-based time series from Alpine meteorological stations

SHAP Attributions (Summed attribution values of the tokens normalized to sum to 1):

- Historical station measurements: 0.45
- Temporal proximity to 2022: 0.30
- Geographical relevance (Alps region): 0.20

Chain-of-Thought Explanation: Our system primarily verifies whether a result includes station-level temperature data for the year 2022 (highest contribution: 0.45). Then it checks whether the temporal coverage aligns with the query year (0.30). Finally, a check for Alpine region specificity (0.20). A small contribution is attributed to metadata quality (0.05).

D. Query-Level Qualitative Analysis with CoT and SHAP Explanations

We further illustrate our approach using two representative sample queries, retrieved from the ENVRI search interface (top 5 results). For each case, we show Chain-of-Thought (CoT) reasoning generated by the LLM, augmented with SHAP

values produced by our reward model to explain the ranking decisions.

Example Query Reasoning: “tsunamis happened in Sicily since 2000”

Top Search Result Types: Reports on the 2002 Stromboli tsunami, tsunami near Palermo

Simulated SHAP Attributions (Summed attribution values of the tokens normalized to sum to 1):

- Mention of known event (e.g., 2002 Stromboli tsunami): 0.50
- Date $\geq$ 2000: 0.30
- Precise geographic mention (Sicily coast): 0.15

Chain-of-Thought Explanation: The system initially detects references to tsunami events in Sicily that occurred after 2000 (most significant factor: 0.50). Followed by it confirms temporal alignment with the query (0.30), and checks for geographic specificity to the Sicilian coast (0.15). In particular, a minor contribution is attributed to source credibility and scientific rigor (0.05).

III. CONCLUSION

Our evaluation depicts in detail how the proposed method combines SHAP attributions and explicit chain-of-thought reasoning to provide interpretable explanations. This helps users gain transparency into why a specific result is ranked higher, not only via a numerical score, but through a clear and logical textual reasoning path grounded by relevant features. This approach enhances trust and facilitates users’ understanding of the ranking beyond a black-box ranking score.

ACKNOWLEDGMENT

The work was made possible through funding from several European Union projects: ENVRI-Hub Next (101131141), EVERSE (101129744), BlueCloud-2026 (101094227), OSCARS (101129751). This research was partially funded by the Dutch Research Council (NWO) Large-Scale Research Infrastructures (LSRI) programme for the LTER-LIFE (<http://www.lter-life.nl>) infrastructure (grant 184.036.014).

REFERENCES

- [1] Siamak Farshidi, Xiaofeng Liao, Na Li, Doron Goldfarb, Barbara Magagna, Markus Stocker, Keith Jeffery, Peter Thijssen, Christian Pichot, Andreas Petzold, et al. Knowledge sharing and discovery across heterogeneous research infrastructures. *Open Research Europe*, 1:68, 2023.
- [2] Jingtong Gao, Xiangyu Zhao, Muiyang Li, Minghao Zhao, Runze Wu, Ruocheng Guo, Yiding Liu, and Dawei Yin. Smlp4rec: an efficient all-mlp architecture for sequential recommendations. *ACM Transactions on Information Systems*, 42(3):1–23, 2024.
- [3] Jonathan L Herlocker, Joseph A Konstan, and John Riedl. Explaining collaborative filtering recommendations. In *Proceedings of the 2000 ACM conference on Computer supported cooperative work*, pages 241–250, 2000.
- [4] Yuhua Li, Xinni Zhang, Linhao Luo, Heng Chang, Yuxiang Ren, Irwin King, and Jia Li. G-refer: Graph retrieval-augmented large language model for explainable recommendation. In *Proceedings of the ACM on Web Conference 2025*, pages 240–251, 2025.